Timing

EXRES
OCHAROMCS
VCHARAMCS
OCHARAMCS
HFLIP
VFLIP
DMACS
DMA
CLR1
CPURES
OCROMDTACK
SVCDTACK
OCRAMDTACK
CLK2
CLK21
CLK22
CPUCLK
SYNCRLDV
VCNTEN
OSCANCLR
HSYNC
VBLANK
CBLK
VSYNC
CSYNC
OBJBUFINIT
DMAINIT
DMACLR
OBJBUFCLR
FBREF
DB0
P1HD
P1HBD
X1SD
WRD
N2HD
P2HD
P4HD
8HD
1VD
{HVCOUNTERS}
P256VD
P256VD
TIMD
TIM1D
TIM1D
TIM2D
TIM2D
TIM3D
RASBD
CASBD
RS1D
RS2D
CS1D
CS2D
OE1D
OE2D
P1H
P1HB
X1S
WR
N2H
P2H
P4H
8H
1V
{HVCOUNTERS}
P256VD
P256VD
TIM
TIM1
TIM1
TIM2
TIM2
TIM3
RASB
CASB
RS1
RS2
CS1
CS2
OE1
OE2

File: timing.kicad_sch

Sprite Table

CLR1
CLK21
OBJBUFCLR
OBJBUFINIT
OSCANCLR
P256VD
DMAINIT
P1HB
P1H
P2H
FBREF
8H
HFLIP
P2HP4H
N2HP4H
DMAON
TIM
TIM3
DMACLR
CHAPAGE
OBJCS
SOBJCS
OCHARAMCS
SREAD
MREAD2
OBJSCRH
OBJSCRV
SLWR
SUWR
MLWR
MUWR
SAB[1..14]
SA[15..20]
AB[1..13]
CRW[0..3]
SPA[0..18]
SPZ17
OCHFD
FBWM1D
FBWM2D
VD[7..0]D
VPD8D
HD[8..2]D
HJ[0..1]D
C[0..3]D
CX[0..3]D
DMAD
SDB[0..15]
DB[0..15]
CLR1
CLK21
OBJBUFCLR
OBJBUFINIT
OSCANCLR
P256VD
DMAINIT
P1HB
P1H
P2H
FBREF
8H
HFLIP
P2HP4H
N2HP4H
DMAON
TIM
TIM3
DMACLR
CHAPAGE
OBJCS
SOBJCS
OCHARAMCS
SREAD
MREAD2
OBJSCRH
OBJSCRV
SLWR
SUWR
MLWR
MUWR
SAB[1..14]
SA[15..20]
AB[1..13]
CRW[0..3]
SPA[0..18]
SPZ17
OCHF
FBWM1
FBWM2
VD[7..0]
VPD8
HD[8..2]
HJ[0..1]
C[0..3]
CX[0..3]
DMA
SDB[0..15]
DB[0..15]

File: sprite_table.kicad_sch

Sprite ROM and RAM

P2H
N2H
P4H
CRW[0..3]
SREAD
OCHARAMCS
OCHAROMCS
VCHARAMCS
SPA[0..18]
SPZ17
SAB1
OCHF
P2HP4H
N2HP4H
VCWENH
VCWENL
VCRDL
VCRDH
SDB[0..15]
W[0..15]
P2H
N2H
P4H
CRW[0..3]
SREAD
OCHARAMCS
OCHAROMCS
VCHARAMCS
SPA[0..18]
SPZ17
SAB1
OCHF
P2HP4H
N2HP4H
VCWENH
VCWENL
VCRDL
VCRDH
SDB[0..15]
W[0..15]

File: sprite_rom_and_ram.kicad_sch

Sprite Frame Buffers

CLR1
CLK21
CLK22
HSYNC
X1S
P2H
N2H
P256VD
P256VD
VCNTEN
WR
HFLIP
VFLIP
W[0..15]
OCD[0..7]
OCD[0..7]
OCHF
FBWM1
FBWM2
VD[0..7]
VPD8
HD[2..8]
HJ[0..1]
C[0..3]
CX[0..3]
CS1
OE1
RS1
OE2
RS2
CS2
CLR1
CLK21
CLK22
HSYNC
X1S
P2H
N2H
P256VD
P256VD
VCNTEN
WR
HFLIP
VFLIP
W[0..15]
OCD[0..7]
OCHF
FBWM1
FBWM2
VD[0..7]
VPD8
HD[2..8]
HJ[0..1]
C[0..3]
CX[0..3]
CS1
OE1
RS1
OE2
RS2
CS2

File: sprite_frame_buffers.kicad_sch

Timing diagrams

File: timing_diagrams.kicad_sch

Sprite timing diagrams

File: sprite_timing_diagrams.kicad_sch

Spare parts on video PWB(B) 302191A

B_G2C 74LS04
B_G2B 74LS04
B_G2E 74LS04
B_G2D 74LS04
B_H11B 74LS74
B_A11A 74LS32

Spare parts are set as NC.

Spare parts on main PWB(A) 302190A

A_L2A 74LS00
A_L2D 74LS00
A_H8A 74LS07
A_H8D 74LS07
A_K1C 74LS32
A_L1C 74LS07
A_L1B 74LS07
A_L8F 74HC367
A_N3B 74LS139
GND

KiCad does not support reference designators starting with a number. Therefore the references used in the original schematics have been mirrored. 6R -> R6.
The original schematics had separate pages for the main and video PCB. They are combined here. This makes some references duplicate. To distinguish between boards A_ is added before part positions on PWB(A) and for PWB(B) B_ and so on. If there are still conflicts a "P" (Position) is added after board letter AP_, BP. Sometimes there are duplicate reference designators then "CAP" maybe added after board letter ACAP_ (CAP is for capacitor).

Tile Layers

EXRES
CLR1
CLK22
TIM1
TIM2
P1HB
X1S
P2H
P4H
P2HP4H
N2HP4H
1V
HFLIP
HSYNC
SYNCRLDV
VCNTEN
RASB
CASB
VCRDL
VCRDH
VCWENL
VCWENH
SYSFLAG
VRAMCS
SVRAMCS
VRAMCVF
VCHARAMCS
V1SCRH
V2SCRH
MLWR
MUWR
MULWR
SLWR
SUWR
MREAD2
SREAD
SA[15..17]
SAB[1..14]
AB[1..13]
1C[0..6]
2C[0..6]
SDB[0..15]
DB[0..15]
EXRES
CLR1
CLK22
TIM1
TIM2
P1HB
X1S
P2H
P4H
P2HP4H
N2HP4H
1V
HFLIP
HSYNC
SYNCRLDV
VCNTEN
RASB
CASB
VCRDL
VCRDH
VCWENL
VCWENH
SYSFLAG
VRAMCS
SVRAMCS
VRAMCVF
VCHARAMCS
V1SCRH
V2SCRH
MLWR
MUWR
MULWR
SLWR
SUWR
MREAD2
SREAD
SA[15..17]
SAB[1..14]
AB[1..13]
1C[0..6]
2C[0..6]
SDB[0..15]
DB[0..15]

File: tile_layers.kicad_sch

Color mixer

CLK2
CBLK
PI[0..2]
FX[0..7]
1C[0..6]
2C[0..6]
OCD[0..7]
COLORRAM
AB[1..12]
MLWR
MREAD1
REDD
GREEND
BLUED
DB[0..7]
CLK2
CBLK
PI[0..2]
FX[0..7]
1C[0..6]
2C[0..6]
OCD[0..7]
COLORRAM
AB[1..12]
MLWR
MREAD1
REDD
GREEND
BLUED
DB[0..7]

File: color_mixer.kicad_sch

IO

VSYNC
SPEAKER+
SPEAKER-
RW
IOCS
CSYNC
MA16
AB[1..4]
DB[0..7]
EXRES
SINT
MINT16EN
DMAON
SOUND-DATA
SOUND-ON
VSYNC
SPEAKER+
SPEAKER-
R-{W
IOCS
CSYNC
RED
GREEN
BLUE
MA16
AB[1..4]
DB[0..7]
EXRES
SINT
MINT16EN
DMAON
SOUND-DATA
SOUND-ON

File: io.kicad_sch

Sound

EXRES
DB[0..7]
REF2V5
PCM1D
PCM2D
FMLD
FMRD
SPEECHD
EXRES
SOUND-DATA
SOUND-ON
DB[0..7]
REF2V5
PCM1D
PCM2D
FMLD
FMRD
SPEECHD

File: sound.kicad_sch

Sound mixer

REF2V5
PCM1
PCM2
FML
FMR
SPEECH
SPEAKER+
SPEAKER-
REF2V5
PCM1
PCM2
FML
FMR
SPEECH
SPEAKER+
SPEAKER-

File: sound_mixer.kicad_sch

Only used on GX903 – Cue Brick

Battery Powered RAM

CPURES
IOCS
MREAD2
MLWR
MUWR
AB[10..1]
DB[15..0]
MA16
CPUCLK
VBLANK
CPURES
IOCS
MREAD2
MLWR
MUWR
AB[10..1]
DB[15..0]
MA16
CPUCLK
VBLANK

File: bpram.kicad_sch

Main CPU

EXRES
CPURES
CPUCLK
TIM1
TIM3
VBLANK
DMAINIT
MINT
MINT16EN
SCPUCLK
HFLIPD
VFLIPD
FIXVRAMCS
VRAMCS
OBJCS
DMACS
COLORRAM
IOCS
OBJSCRH
OBJSCRV
V1SCRH
V2SCRH
PI[0..2]
VRAMCVF
SYSFLAG
MCOMCS
RW
MREAD1
MREAD2
MLWR
MUWR
MULWR
MA[1..23]
AB[1..13]
DB[0..15]
SCPUCLK
HFLIP
VFLIP
FIXVRAMCS
VRAMCS
OBJCS
DMACS
COLORRAM
IOCS
OBJSCRH
OBJSCRV
V1SCRH
V2SCRH
PI[0..2]
VRAMCVF
SYSFLAG
MCOMCS
RW
MREAD1
MREAD2
MLWR
MUWR
MULWR
MA[1..23]
AB[1..13]
DB[0..15]
MA16

File: mcpu.kicad_sch

Shared Resources

CPUCLK
MCOMCS
MREAD1
MLWR
MUWR
SCOMCS
SREAD
SLWR
SUWR
AB[1..13]
SAB[1..13]
DB[0..15]
SDB[0..15]
CPUCLK
MCOMCS
MREAD1
MLWR
MUWR
SCOMCS
SREAD
SLWR
SUWR
AB[1..13]
SAB[1..13]
DB[0..15]
SDB[0..15]

File: shared_resources.kicad_sch

Slave CPU

EXRES
CPURES
DMAINIT
SCPUCLK
TIM2
VBLANK
SINT
TIM3
OCRAMDTACK
OCROMDTACK
SVCDTACK
CHAPAGED
MINTD
SOBJCSD
SVRAMCSD
SCOMCSD
OCHARAMCSD
OCHAROMCSD
VCHARAMCSD
SREADD
SLWRD
SUWRD
SAB[1..14]
SDB[0..15]
SA[15..23]
EXRES
CPURES
DMAINIT
SCPUCLK
TIM2
VBLANK
SINT
TIM3
OCRAMDTACK
OCROMDTACK
SVCDTACK
CHAPAGE
MINT
SOBJCS
SVRAMCS
SCOMCS
OCHARAMCS
OCHAROMCS
VCHARAMCS
SREAD
SLWR
SUWR
SAB[1..14]
SDB[0..15]
SA[15..23]

File: scpu.kicad_sch

Fixed Layer

CLK21
{HVCOUNTERS}
P1HB
P2H
P4H
HFLIP
FIXVRAMCS
MLWR
MUWR
MREAD2
AB[1..13]
DB[0..15]
FX[0..7]
CLK21
{HVCOUNTERS}
P1HB
P2H
P4H
HFLIP
FIXVRAMCS
MLWR
MUWR
MREAD2
AB[1..13]
DB[0..15]
FX[0..7]

File: fixed_layer.kicad_sch

There are several games and variations on the twin 16 platform:
– Main Board PWB(A)
– Video Board PWB(B)
– Additional ROM Board, if not ROM's mounted directly on video board.
– Battery powered RAM board

	Id	PWB(A)	PWB(B)	PWB(C)
Dark Adventure	GX687	302190A	302191A	–
Gradius 2	GX785	302190A	302191A	–
Vulcan Venture	GX785	302190A	302191A	–
Gradius 2	GX785	302190B	302191B	402145A
Hard Puncher	GX870	302190B	302191B	402145A
M.I.A	GX808	302190B	302191B	402145A
Cue Brick	GX903	302190B	302191B	–

Cue Brick has an additional Battery Powered RAM Board:
PWB451803

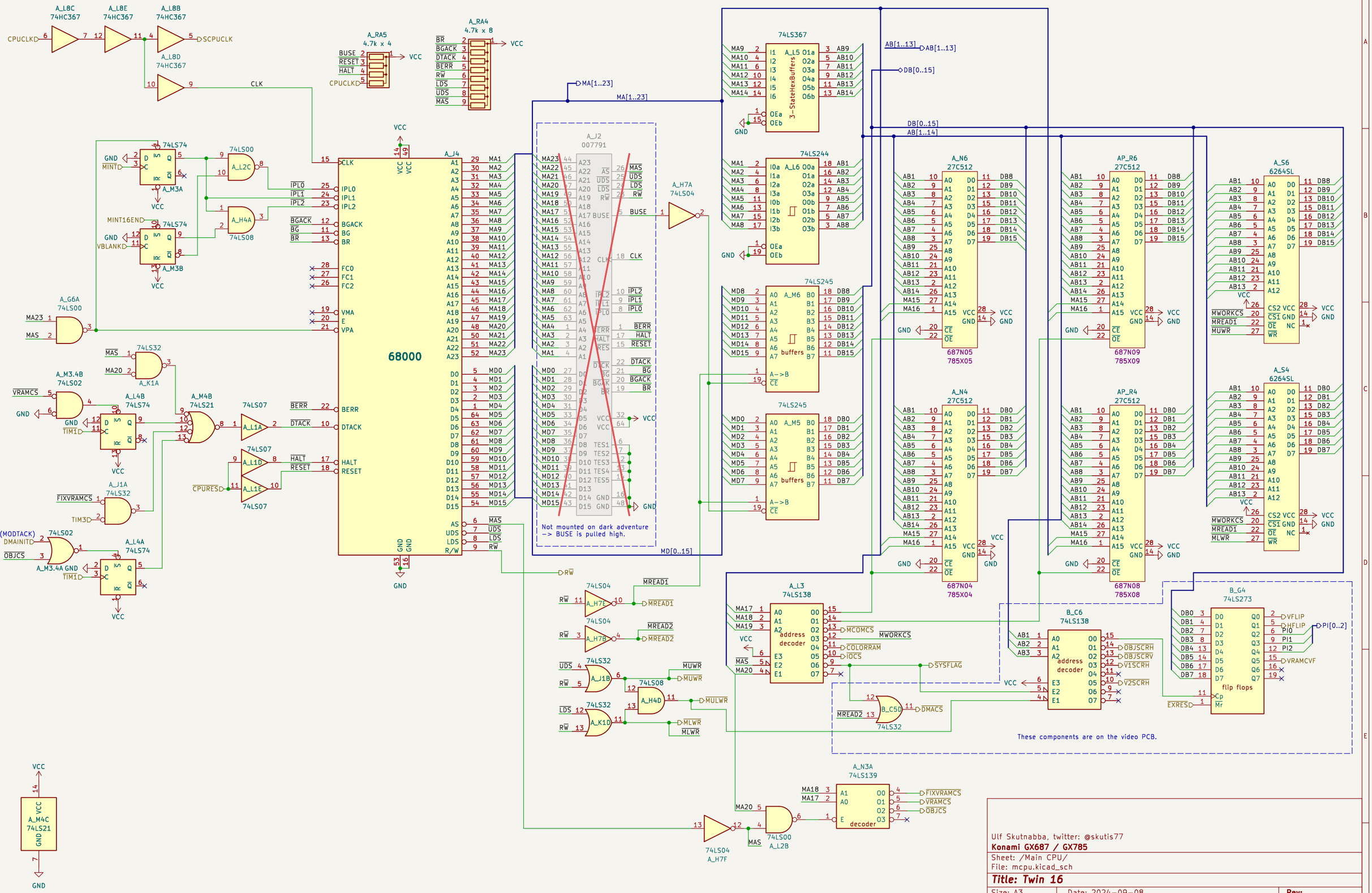
Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /
File: darkadv.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	Rev:
KiCad E.D.A. 8.0.5		Id: 1/17


A_L8C 74HC367
A_L8E 74HC367
A_L8B 74HC367
A_L8D 74HC367
CPUCLKD
SCPUCLK
CLK
A_RA5 4.7k x 4
BUSE
RESET
HALT
CPUCLKD
VCC
A_RA4 4.7k x 8
BR
BGACK
DTACK
BERR
RW
LDS
UDS
MAS
VCC
MA[1..23]
74LS367
3-StateHexBuffers
AB[1..13]
DB[0..15]
74LS244
buffers
74LS245
A->B
DB[0..15]
AB[1..14]
A_N6 27C512
AP_R6 27C512
687N05 785X05
A_S6 6264SL
A_N4 27C512
AP_R4 27C512
687N04 785X04
687N08 785X08
A_S4 6264SL
MWORKCS
MREAD1
MLWR
A_J4
A_J2 007791
Not mounted on dark adventure -> BUSE is pulled high.
MD[0..15]
68000
74LS74
A_M3A
A_M3B
MINTD
MINT16END
VBLANKD
74LS00
A_L2C
A_H4A
74LS08
IPL0
IPL1
IPL2
BGACK
BG
BR
FC0
FC1
FC2
VMA
E
VPA
BERR
DTACK
HALT
RESET
A_H7A 74LS04
A_G6A 74LS00
MAS
MA20
A_K1A
A_M3.4B 74LS02
VRAMCS
A_L4B 74LS74
TIM1D
A_M4B 74LS21
74LS07
A_L1A
A_L1D
A_L1E
CPURES
A_J1A 74LS32
FIXVRAMCS
TIM3D
(MODTACK)
DMAINITD
OBJCS
A_M3.4A
A_L4A 74LS74
A_M4C 74LS21
A_H7E
A_H7B
MREAD1
MREAD2
A_J1B
A_H4D
MUWR
MULWR
A_K1D
MLWR
A_L3 74LS138
address decoder
MCOMCS
COLORRAM
IOCS
SYSFLAG
B_C5D
DMACS
B_C6 74LS138
OBJSCRH
OBJSCRV
V1SCRH
V2SCRH
B_G4 74LS273
flip flops
VFLIP
HFLIP
PI0
PI1
PI2
VRAMCVF
EXRES
PI[0..2]
These components are on the video PCB.
A_N3A 74LS139
FIXVRAMCS
VRAMCS
OBJCS
A_L2B
Ulf Skutnabba, twitter: @skutis77
Konami GX687 / GX785
Sheet: /Main CPU/
File: mcpu.kicad_sch
Title: Twin 16
Size: A3
Date: 2024-09-08
Rev:
KiCad E.D.A. 8.0.5
Id: 2/17

Only used on PWB(A)302190B
Goes to CN2:A23.
On PWB(A)302190A CN2:A23 is connected to GND.

In the original schematics
DMAINT, MODTACK and DMAINIT are the same

8R and 10R are not mounted when
the GX687 board is used for gradius 2
with GX785 roms.

CPU address
in bold

called ZIP RAM
in test screens

Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /Slave CPU/
File: scpu.kicad_sch

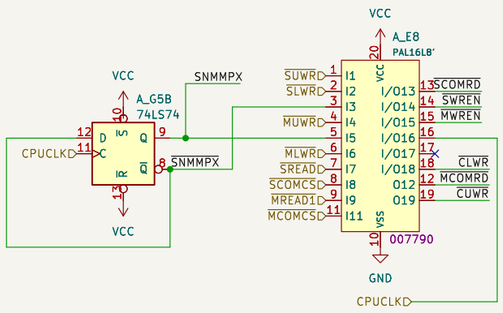
Title: Twin 16

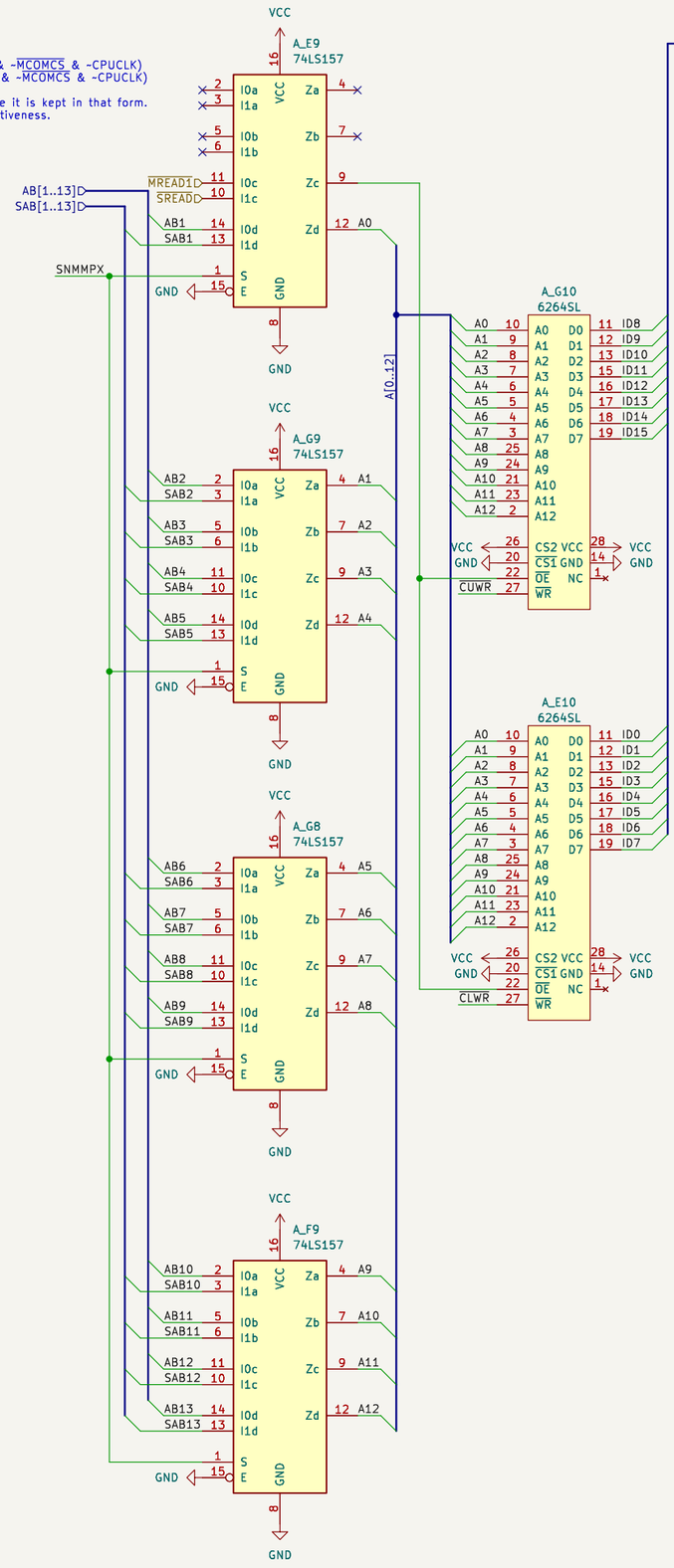
Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 3/17

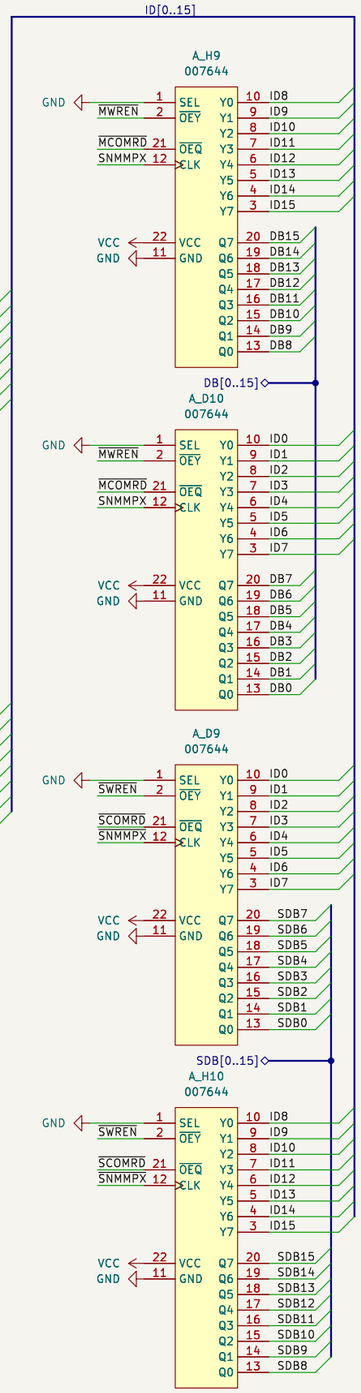
Konami 007790

$\overline{MCOMRD}$ = ~(~$\overline{MREAD1}$ & ~$\overline{MCOMCS}$) = $\overline{MREAD1}$ | $\overline{MCOMCS}$
$\overline{SCOMRD}$ = ~(~$\overline{SREAD}$ & ~$\overline{SCOMCS}$)
$\overline{SWREN}$ = ~(~SNMMPX & $\overline{SREAD}$ & ~$\overline{SCOMCS}$)
$\overline{MWREN}$ = ~(~SNMMPX & $\overline{MREAD1}$ & ~$\overline{MCOMCS}$)
$\overline{CLWR}$ = ~(~$\overline{SLWR}$ & ~SNMMPX & SNMMPX & $\overline{SREAD}$ & ~$\overline{SCOMCS}$ & ~CPUCLK | ~SNMMPX & ~$\overline{MLWR}$ & $\overline{MREAD1}$ & ~$\overline{MCOMCS}$ & ~CPUCLK)
$\overline{CUWR}$ = ~(~$\overline{SUWR}$ & ~SNMMPX & SNMMPX & $\overline{SREAD}$ & ~$\overline{SCOMCS}$ & ~CPUCLK | ~$\overline{MUWR}$ & ~SNMMPX & $\overline{MREAD1}$ & ~$\overline{MCOMCS}$ & ~CPUCLK)

All AND logic can be converted to OR form. But since AND logic and inverters are used in the PAL16L8 device it is kept in that form.
The equations follow verilog syntax. The active low signals are not inverted, the bar is there only to show activeness.







This is from jammarcade.net

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs
mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs.
If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs.
If SEL is LOW then the outputs will only change state on a rising edge CLK.

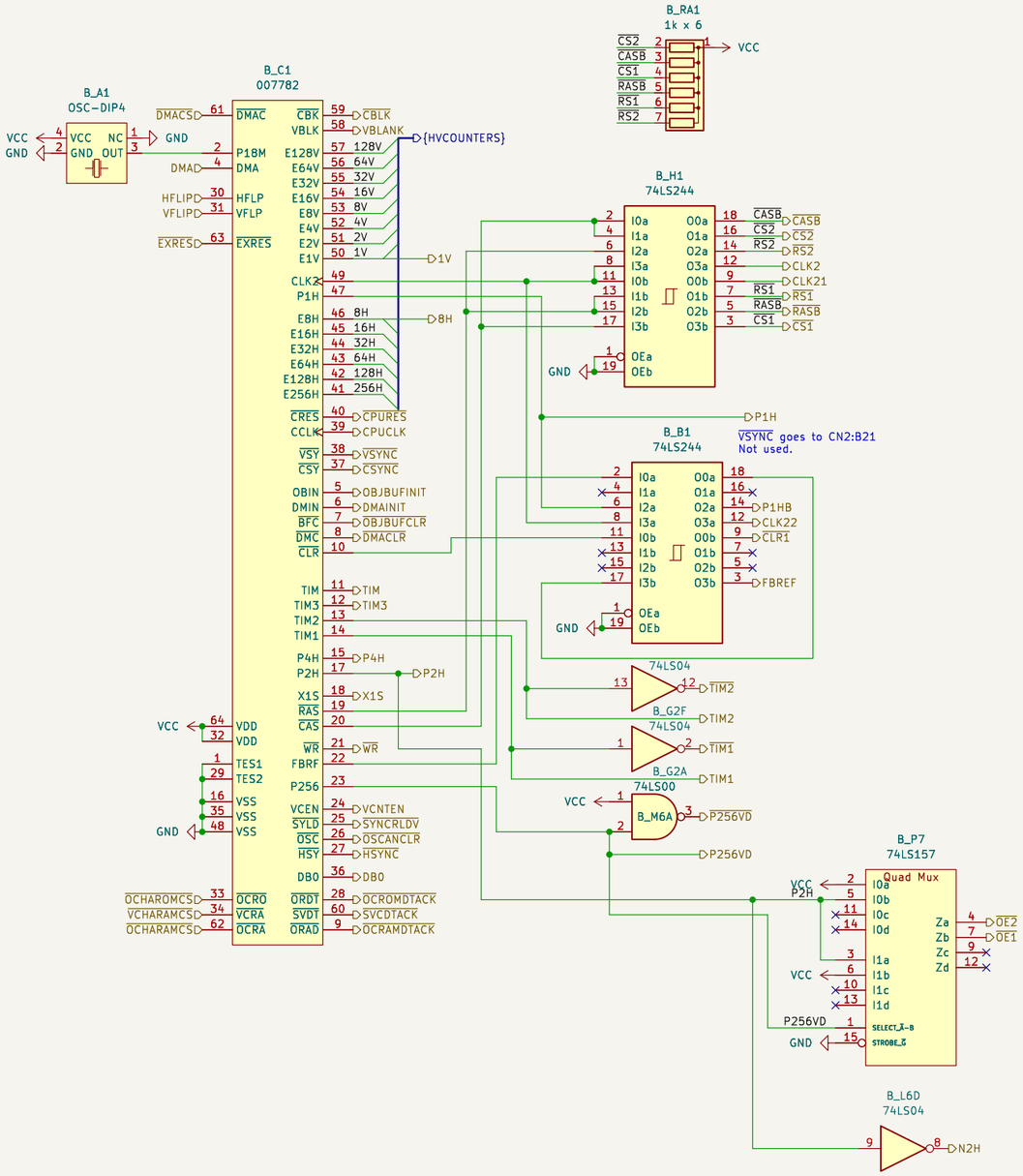
Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

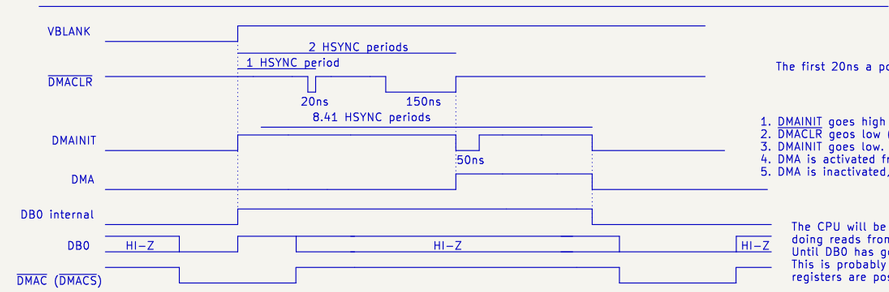
Sheet: /Shared Resources/
File: shared_resources.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 4/17



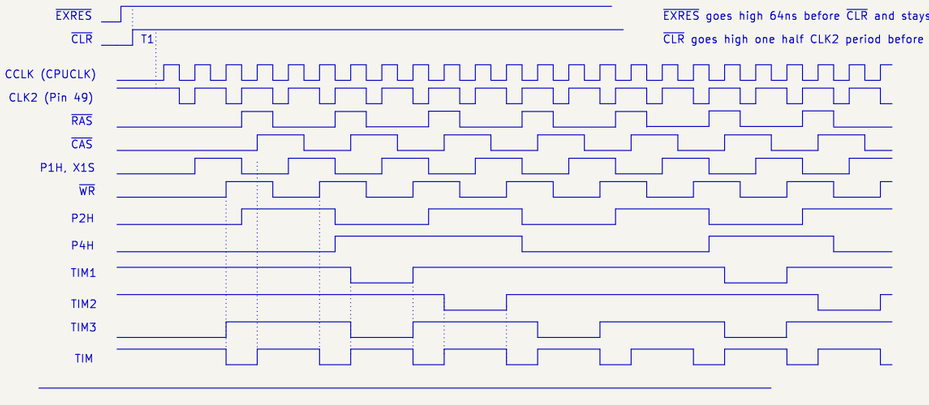
DMA Synchronization



The first 20ns a possible glitch?

1. DMAINIT goes high after VBLANK goes active
2. DMACLR geos low (the second time) and high after 150ns.
3. DMAINIT goes low.
4. DMA is activated from the 007783(Sprite engine).
5. DMA is inactivated, DMAINIT, DBO internal goes low.

The CPU will be activating regularly DMACS,
doing reads from the system registers periodically.
Until DBO has gone from low to high and low again.
This is probably to see if writes to the sprite controller
registers are possible.



EXRES goes high 64ns before CLR and stays high.

CLR goes high one half CLK2 period before the first falling edge of CLK2.

CCLK = 18.432MHz / 2 = 9.216MHz
CLK2 = 18.432MHz / 3 = 6.144MHz,
T = 1/f = 162.76ns, T = 2 x T1

f = CLK2 / 2 = 3.072MHz, T2 = 3 x T1

f = CLK2 / 2 = 3.072MHz

f = CLK2 / 2 = 3.072MHz

f = CLK2 / 4 = 1.536MHz

f = CLK2 / 8 = 768kHz

f = CLK2 / 8 = 768kHz

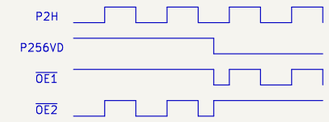
f = CLK2 / 8 = 768kHz

f = CLK2 / 4 = 1.536MHz

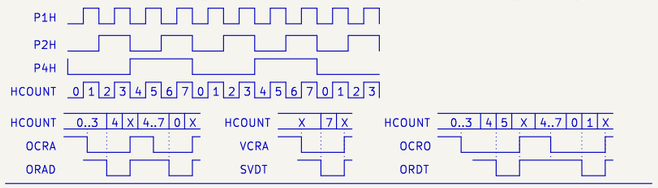
f = CLK2 / 2 = 3.072MHz

Timer	Usage
TIM1	Main CPU Tilemap RAM access
TIM2	Slave CPU RAM Tilemap RAM access
TIM	Sprite timing/address generator 007783
TIM3	Sprite timing/address generator 007783

Output enables active vs. part of frame buffer active



Chip select signals and Data Acknowledge signals



Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /Timing/
File: timing.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 5/17

B_A2
74LS157
Quad Mux
AB[1..13]
AB12
AB13
GND
P2H
B_B2
74LS157
AB8
AB9
AB10
AB11
16V
32V
64V
128V
B_B4B
74LS27
B_L6A
74LS04
FIXVRAMCS
MUWR
B_C2
74LS157
AB4
AB5
AB6
AB7
64H
128H
256H
8V
A[0..12]
B_B4A
74LS27
B_L6F
74LS04
MLWR
B_D2
74LS157
MREAD2
8H
16H
32H
10ns delay
50ns delay from OE active
to output data valid
B_A3
6264SL
B_B3
6264SL
D[0..15]
7ns delay
B_J1A
74LS86
P4H
P4H
20ns delay
B_C4
74LS174
B_C3
74LS174
hex flip flop
Palette, HFLIP and VFLIP is valid for
eight pixels at a time and becomes valid
on the rising edge of P4H
7ns delay
74LS86
B_J1C
P1HB
P1HB
B_J2A
B_J1B
B_J2C
B_J2D
B_H2C
B_H2B
B_H2D
HFLIP
B_D3
74LS174
B_D4
74LS174
RAM
A[0..5] Horizontal position 0..504 in steps of 8
A[6..10] Vertical position 0..248 in steps of 8
A[11.12] Tilemaps? Up to 4.
B_F3
27C128
687M14/L14/N14
785H14/G14
ROM
A[0..1] Horizontal position 0..6 in steps of 2
A[2..4] Vertical position 0..7
A[5..13] Tile number
Two pixels color data is output
on each rising edge of P1HB.
The current pixel data is multiplexed
with P1HB (and HFLIP).
B_E2
74LS273
FX[20..23]
FX[10..13]
B_F2
74LS157
B_J1D
B_H2A
The total delay for a pixel is seven pixel clocks
from when P2H and OE goes high and low respectively.
B_G3
74LS273
B_H3
74LS273
B_H4
74LS273
flip flops
CLK21
FX[0..7]
FX[0..3]: Color
FX[4..7]: Palette
B_D5
74LS245
B_E5
74LS245
buffers
A->B
CE
{HVCOUNTERS}
DB[0..15]
B_C5A
74LS32
D[0..8]: Tile number
D[9..12]: Palette
D13: HFLIP
D14: VFLIP
4. DFF's data
out valid
6. DFF's data
out valid
3. Data out latched
into DFF's.
CLK21
P1H
P1HB
P1HB
P2H
P4H
P4H
1
2
3
4
5
6
7
8
2. Data out from
RAM valid
4. FX[0..7] Data out valid
5. Palette and color values
latched into DFF's 3C and 2E.
1. Addresses and OE
latched into RAM. This condition sets the start:
P2H & P4H = 1.

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Konami GX687 / GX785

Sheet: /Fixed Layer/
File: fixed_layer.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 6/17

ZIP RAM

256kB tile RAM
16-bit access by CPU
32-bit access by video

64k x 4bit

This is from jammarcade.net

Konami 007644

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs. If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs.
If SEL is LOW then the outputs will only change state on a rising edge CLK.

The 007781 is a shift register. All counters related to all timing signals are inside the chip. It is synchronized on the rising edge of CLR and CLK2 after that. It reads 32 bits / 4 bits / pixel = 8 pixels at a time.

1C[0:3]: Layer 1 Pixel color code
1C[4:6]: Layer 1 Palette
2C[0:3]: Layer 2 Pixel color code
2C[4:6]: Layer 2 Palette

CPUCTL flag must be set in 007779 to get the VA bus driven by the CPUs

VD[0..12]: Tile number
VD[13..15]: Palette Color

Data latched from the RAM:
Edge Tilemap
T1–T2 1
T2–T3 2

It is output 2 pixel clocks later to the ROMs.

When VA is switched to output the bus is updated for each P1H cycle.

The vertical counters are not the same value when they are offset with the scroll registers.

Tilemap 1 and 2
VA[5..0]: Horizontal position: 0..63 –> 0..63 x 8 = 0..504
VA[11..6]: Vertical position: 0..63 –> 0..63 x 8 = 0..504
VA12 = 0: Tilemap 1, VA12 = 1: Tilemap 2

Outputs of RAM active when P4H is low, i.e during T1 and T2.

SYNCRLDV: Usually 60.604Hz
T is ~150ns.

VCNTEN: 8.0000kHz

HSYNC: 16.0000kHz, T2: 5.2175us

Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /Tile Layers/
File: tile_layers.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 7/17

B_K1
007783

B_L3
6264SL

B_M3
6264SL

B_K3
007784

74LS245 B_J5
74LS245 B_G5
74LS245 B_H5
74LS245 B_F5

B_K5
007785

74LS157 Quad Mux B_K7
74LS157 Quad Mux B_L7
74LS157 Quad Mux B_N6
74LS157 Quad Mux B_M7
74LS157 Quad Mux B_N7

B_C5B 74LS32
B_J8A 74LS32
B_K6A 74LS32
B_K6C 74LS32

B_H11A 74LS74

B_P1 PAL16L8'

B_RA2 2.2k x 8
B_RA3 2.2k x 8

AB[1..13]
SAB[1..14]
OA[0..12]
CRW[0..3]
OD[0..15]
DB[0..15]
SDB[0..15]
SA[15..20]
O[0..19]
VD[7..0]
HD[8..2]
HJ[0..1]
CX[0..3]
C[0..3]
SPA[0..17]
VPD8
VCC
GND
{RW1}
{RW2}

Each unparsed sprite (object) occupies 40 words in object RAM. Only the first 8 words of each object data is used. The memory range for unparsed data is up 17FFh. That leaves room for 153 (153 x 40 −1 = 17E7h) objects per frame.

On PWB(A) 302190A CHAPAGE (CN2:A23) is connected to GND

007784 Sprite parser

− subtracts the scroll value for the sprites from current coordinates for the sprites.
− When DMA goes high. The 007784 will read 8 words, if bit OD15 of the first word is high then BEFLAG goes high and stays high until the sprite RAM has processed the first four words and possibly updated the four following words at the high end of the RAM. It will compress the eigth words into four words and also subtract the x− and y−scroll values from the x and y coordinates. It will write the four words at the high end of the sprite RAM.
− Addresses at the low end goes from 0x0000..0x0007 and the following cycle at 0x0028..0x002F..

Scroll Register
OBJSCRV Sprite horizontal scroll
OBJSCRH Sprite vertical scroll

Data is latched in the scroll registers when the signals OBJSCRV and OBJSCRH go low.

007793

CLK21′ = CLK21

LRW = ~(DMAONS & OBJBUFINIT & CLK21′ | BEFLAG & DMA & P1H & DMAONS & CLK21′ & (~8H & HFLIP | 8H & ~HFLIP) ~LWR & ~DMAINIT)

URW = ~(DMAONS & OBJBUFINIT & CLK21′ | BEFLAG & DMA & P1H & DMAONS & CLK21′ & (8H & ~HFLIP | ~8H & HFLIP) | ~UWR & ~DMAINIT)

The sprite memory is controlled in three separate cases:
1. OBJBUFINIT
2. BEFLAG & DMA
3. ~DMAINIT

Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /Sprite Table/
File: sprite_table.kicad_sch

Title: Twin 16

Size: A3	Date: 2024−09−08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 8/17

Graphics Tile Data (ROM and RAM)

This is from jammarcade.net

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs. If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs. If SEL is LOW then the outputs will only change state on a rising edge CLK.

Konami 007788

The inverted ~P2H is written as N2H.

$\overline{OCRDL}$ = ~(~$\overline{SREAD}$ & SAB1 & ~$\overline{OCHARAMCS}$ & $\overline{OCHAROMCS}$ | ~$\overline{SREAD}$ & SAB1 & ~$\overline{OCHAROMCS}$)
$\overline{OCWL}$ = ~(~P2H & $\overline{SREAD}$ & SAB1 & ~$\overline{OCHARAMCS}$)
$\overline{OCRDH}$ = ~(~$\overline{SREAD}$ & ~SAB1 & ~$\overline{OCHARAMCS}$ & $\overline{OCHAROMCS}$ | ~$\overline{SREAD}$ & ~SAB1 & ~$\overline{OCHAROMCS}$)
$\overline{OCWH}$ = ~(~P2H & $\overline{SREAD}$ & ~SAB1 & ~$\overline{OCHARAMCS}$)
$\overline{VCWENL}$ = ~(P2H & P4H & ~$\overline{VCHARAMCS}$ & $\overline{SREAD}$ & SAB1)
$\overline{VCRDL}$ = ~(~$\overline{VCHARAMCS}$ & ~$\overline{SREAD}$ & SAB1)
$\overline{VCRDH}$ = ~(~$\overline{VCHARAMCS}$ & ~$\overline{SREAD}$ & ~SAB1)
$\overline{VCWENH}$ = ~(P2H & P4H & ~$\overline{VCHARAMCS}$ & $\overline{SREAD}$ & ~SAB1)
$\overline{P2HP4H}$ = ~(P2H & P4H)
$\overline{N2HP4H}$ = ~(~P2H & P4H)

B_S7 007644
B_T8 007644
B_R7 007644
B_U8 007644
B_J6 GAL22V10 007788
B_M6C 74LS00
B_L6B 74LS04
B_M6B 74LS00
23C4100 B_L10
23C4100 B_P10
23C4100 B_M10
687I15 785F15
687I17 785F17
687I16 785F16
687I18 785F18
Vertical flipped ROM's?
B_M8 SRAM32K8B
B_L8 SRAM32K8B
B_R8 SRAM32K8B
B_P8 SRAM32K8B
128kB RAM for dynamic tiles
74LS86 B_J2B
B_T9 74LS374
B_T10 74LS374
B_U9 74LS374
B_U10 74LS374
74LS04 B_L6C
3-state flip flops

Pixel color data output rate is:
16-bits / 2 pixels = 8 bits / pixel.

W[15]: Vertical flip

Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /Sprite ROM and RAM/
File: sprite_rom_and_ram.kicad_sch

Title: Twin 16

Size: A3 | Date: 2024-09-08 | Rev:
KiCad E.D.A. 8.0.5 | Id: 9/17

Frame buffer 1

Frame buffer 2

2 x 256kB frame buffers
Framebuffer 1 is outputting pixel data when P256VD is low and writes to the FB is possible when P256VD is high. The conditions are reveresed for FB2.

Not certain which of the parts 6KB,6UB and 6UD are used where. Some errors in the original schematics.

Writes to FB are possible when the "pixel part" is sleected from the mux.
If this expression is true
(R[0:3] != 4'h0) && (FBWM1 | FBWM2 | -WR | -VPD8)
data is written to DRAM.

Data is written when:
– WR is active.
– Not a transparent pixel (4'h0).
– FBWM1 and FBWM2 are not active: No writes to frame buffer going on.
– VPD8 (vertical counter highest bit, VCNT > 256) is not active. Outside visible area.

OCD[0:3]: Pixel color
OCD[4:7]: Palette

Frame buffer 1 Palette color

Frame buffer 2 Palette color

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Konami GX687 / GX785

Sheet: /Sprite Frame Buffers/
File: sprite_frame_buffers.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	Rev:
KiCad E.D.A. 8.0.5		Id: 10/17

Components are on Video PCB

Components are on Main PCB

object layer:
0000: Transparent
1111: Shadow
else: Opaque

B_A11D 74LS32 · B_A11B 74LS32 · B_A11C 74LS32 · B_J8B 74LS32 · B_J8C 74LS32 · B_J8D 74LS32

B_J11 GAL22V10 007789

PI[0..2] · {SEL} · FX · 1C · SHADOW · SELB · SELA · 2C3 · 2C2 · 2C1

B_H9 74LS153 Mux 4–1 → COLOR8 / COLOR9

B_H8 74LS153 Mux 4–1 → COLOR6 / COLOR7

B_H10 74LS153 Mux 4–1 → COLOR4 / COLOR5

B_J9 74LS153 Mux 4–1 → COLOR2 / COLOR3

B_J10 74LS153 Mux 4–1 → COLOR0 / COLOR1

2C[0..6] · 1C[0..6] · OCD[0..7] · FX[0..7]

COLOR[0..9]

```
Konami 007789

i11 (1C6) and i5 (PI2) is not used

~o15 = OCD0 & OCD1 & OCD2 & OCD3 | ~OCD0 & ~OCD1 & ~OCD2 & ~OCD3

o16 = OCD0 & OCD1 & OCD2 & OCD3

SELA = FX +
       /PI1 &   PI0 & 1C +
       /PI1 &  OCD0 &  OCD1 &  OCD2 &  OCD3 & 1C +
       /PI1 & /OCD0 & /OCD1 & /OCD2 & /OCD3 & 1C +
        PI1 &  OCD0 &  OCD1 &  OCD2 &  OCD3 & /2C0 & /2C1 & /2C2 & /2C3 +
        PI1 & /OCD0 & /OCD1 & /OCD2 & /OCD3 & /2C0 & /2C1 & /2C2 & /2C3

SELB =/FX & OCD0 & OCD1 & OCD2 & OCD3 +
       /FX & /OCD0 & /OCD1 & /OCD2 & /OCD3 +
       /FX & PI0 & /PI1 & 1C +
       /FX & PI0 &  PI1 & 2C6 & 2C0 +
       /FX & PI0 &  PI1 & 2C6 & 2C1 +
       /FX & PI0 &  PI1 & 2C6 & 2C2 +
       /FX & PI0 &  PI1 & 2C6 & 2C3
o21.oe = vcc

/SHADOW = /FX & /PI1 & OCD0 & OCD1 & OCD2 & OCD3 +
          /FX & /PI0 & OCD0 & OCD1 & OCD2 & OCD3 & /1C  +
          /FX &  PI0 & OCD0 & OCD1 & OCD2 & OCD3 & /2C6 +
          /FX &  PI0 & OCD0 & OCD1 & OCD2 & OCD3 & /2C0 & /2C1 & /2C2 & /2C3
```

A_U6 74LS273 (flip flops) · CLK2 · COLOR2–COLOR9 → CB2–CB6

A_T6 74LS157 (Quad Mux) · AB9 · AB10 · AB11 · AB12 · COLORRAM · SELECT_A–B · STROBE_G → C9 · C8 · C7 · C6

AB[1..12] · CB[0..6] · CBLK

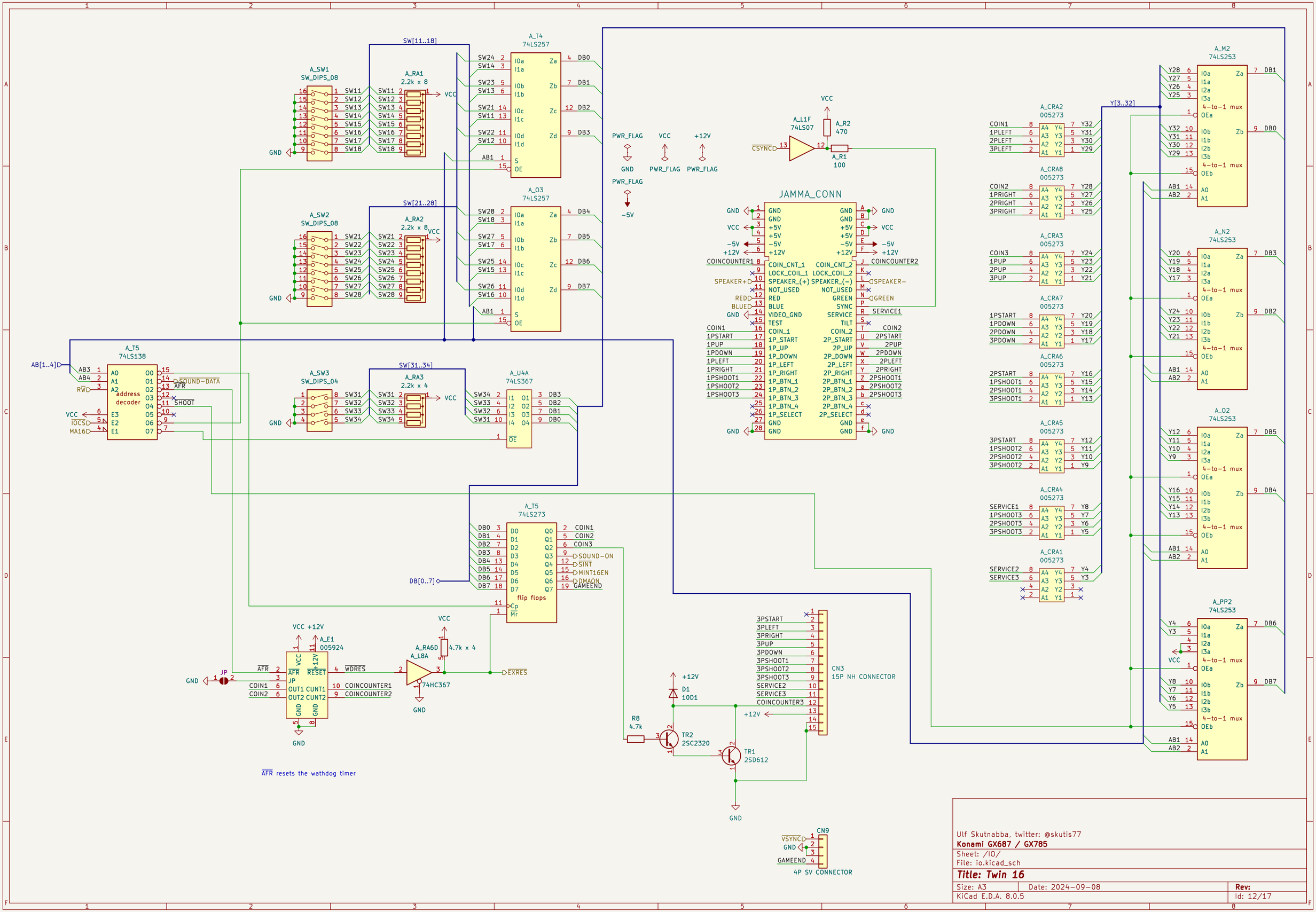
A_U7 74LS174 (hex flip flop) · CBLK · COLOR0 · COLOR1 → CB0 · CB1

A_F7B 74LS04 · AB1

A_U1 007327 · CB[6..9] · DB[0..7] · CLK2 · MREAD1 · COLORRAM · MLWR

R 82 → RED · A_R4 82 → GREEN · A_R5 82 → BLUE

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Konami GX687 / GX785	
Sheet: /Color mixer/	
File: color_mixer.kicad_sch	
Title: Twin 16	
Size: A3	Date: 2024-09-08
KiCad E.D.A. 8.0.5	Id: 11/17



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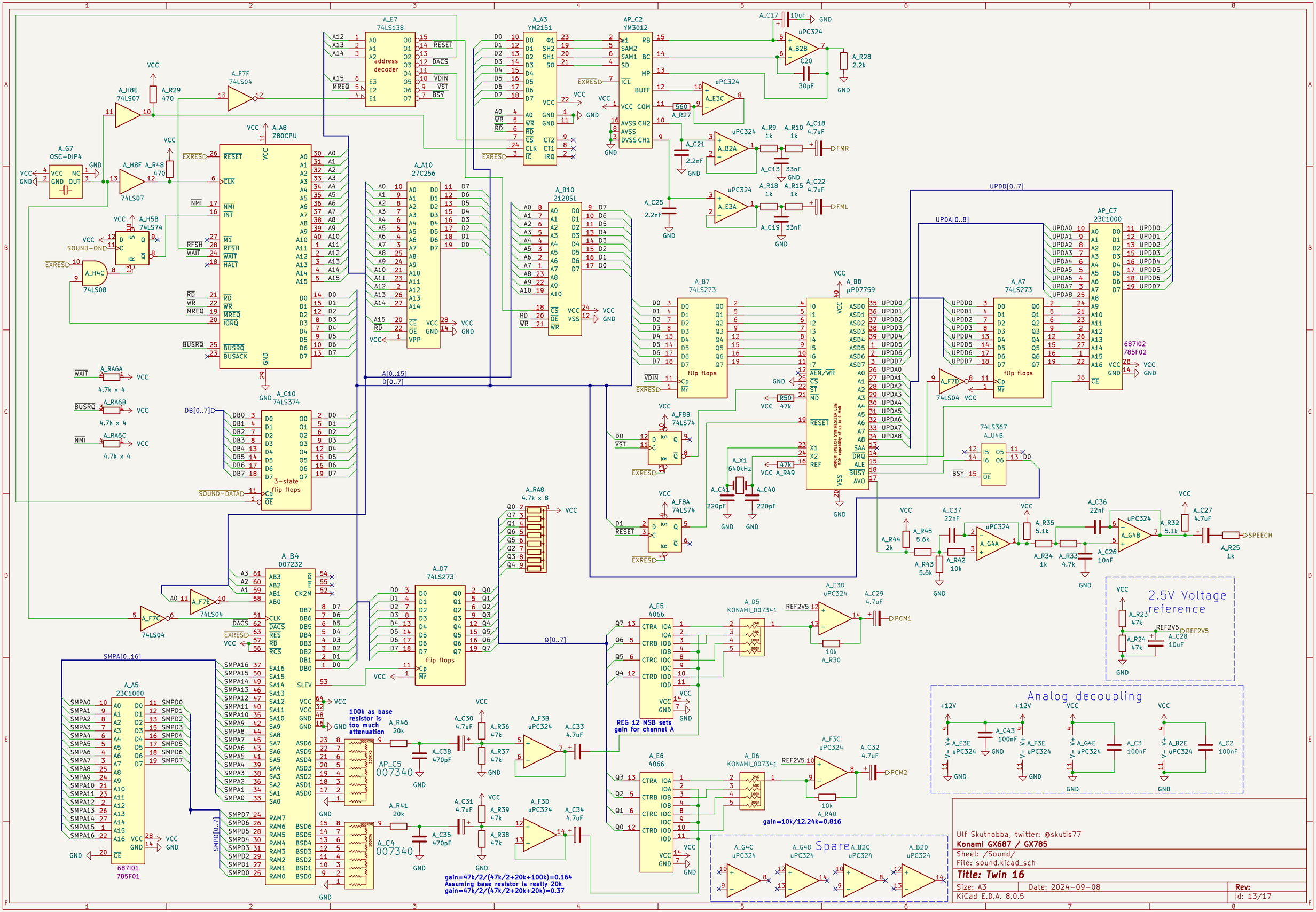
Konami GX687 / GX785

Sheet: /IO/

File: io.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 12/17


A_E7
74LS138
address decoder
A_A3
YM2151
AP_C2
YM3012
A_F7F
74LS04
A_H8E
74LS07
A_R29
470
A_G7
OSC-DIP4
A_H8F
74LS07
A_R48
470
A_A8
Z80CPU
A_A10
27C256
A_B10
2128SL
A_H5B
74LS74
A_H4C
74LS08
A_B7
74LS273
A_B8
µPD7759
ADPCM SPEECH SYNTHESIZER LSI
ROM capability of up to 1 Mbit
A_A7
74LS273
AP_C7
23C1000
687I02
785F02
A_F7D
74LS04
74LS367
A_U4B
A_C10
74LS374
3-state
flip flops
A_F8B
74LS74
A_F8A
74LS74
A_X1
640kHz
A_RA6A
4.7k x 4
A_RA6B
4.7k x 4
A_RA6C
4.7k x 4
A_RA8
4.7k x 8
A_B4
007232
A_D7
74LS273
A_E5
4066
A_E6
4066
A_D5
KONAMI_007341
A_D6
KONAMI_007341
AP_C5
007340
A_C4
007340
A_A5
23C1000
687I01
785F01
A_F7E
74LS04
A_F7C
74LS04
A_E3D
uPC324
A_E3C
uPC324
A_B2A
uPC324
A_E3A
uPC324
A_B2B
uPC324
A_G4A
uPC324
A_G4B
uPC324
A_F3B
uPC324
A_F3D
uPC324
A_F3C
uPC324
A_G4C
uPC324
A_G4D
uPC324
A_B2C
uPC324
A_B2D
uPC324
Spare
2.5V Voltage reference
Analog decoupling
100k as base resistor is too much attenuation
REG 12 MSB sets gain for channel A
gain=10k/12.24k=0.816
gain=47k/2/(47k/2+20k+100k)=0.164
Assuming base resistor is really 20k
gain=47k/2/(47k/2+20k+20k)=0.37
A[0..15]
D[0..7]
Q[0..7]
SMPA[0..16]
SMPD[0..7]
DB[0..7]
UPDA[0..8]
UPDD[0..7]
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Konami GX687 / GX785
Sheet: /Sound/
File: sound.kicad_sch
Title: Twin 16
Size: A3
Date: 2024-09-08
KiCad E.D.A. 8.0.5
Rev:
Id: 13/17

To enable stereo sound output on connector CN8:
Use the connector which shortens Pin 1 -> Pin 2 and
Pin 3 -> Pin 4 on Connector CN7.

FMRD
SPEECHD
PCM1D
PCM2D
FMLD
SPEECHD
PCM1D
PCM2D
REF2V5D
REF2V5D
A_R17 1k
A_R12 7.5k
A_R11 5.1k
A_R19 5.1k
A_R16 1k
A_R22 7.5k
A_R31 5.1k
A_R20 5.1k
A_E3B uPC324
A_F3A uPC324
A_R13 12k
A_R21 12k
+12V
+12V
A_R52 3.3k
A_R53 4.7uF
A_C15 4.7uF
A_C16
A_R14 10k
A_R26 10k
CN6 4P NH CONN
CN7 4P NH CONN
GND
GND
APOT_R53 1k
APOT_R52 1k
A_C11 100nF
A_C12 100nF
GND
GND
GND
GND
+12V
A_C9 100uF
A_D1 MB3722
ACAP_C3 1000uF
GND
GND
ACAP_C10 100uF
A_C14 10uF
GND
GND
IN-A
IN+A
IN+A
IN-B
VCC
GNDIN
GNDOUT
BYPASS
BSA
OUTA
BSB
OUTB
A_C8 100uF
A_C7 100uF
ACAP_C2 1000uF
A_C6 100nF
A_R6 1
A_C5 100nF
A_R7 1
A_C1 1000uF
GND
GND
GND
SPEAKER-
SPEAKER+
CN8 4P NH CONN

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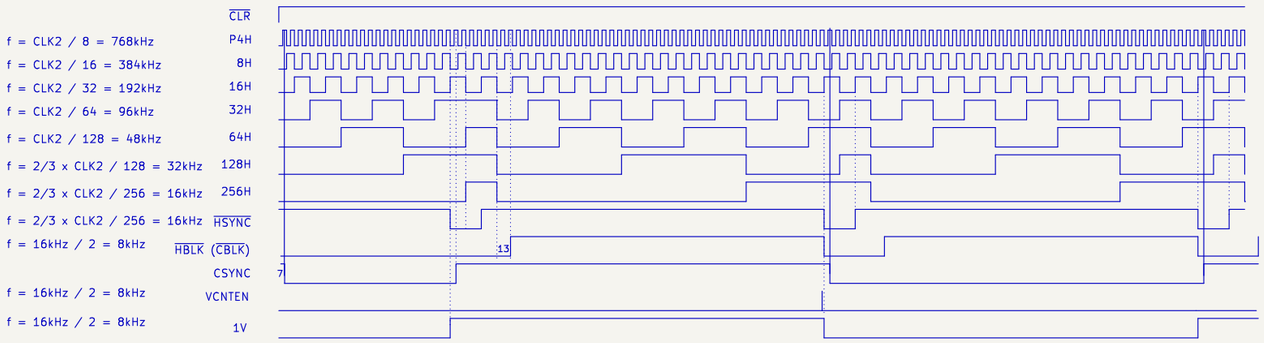
Konami GX687 / GX785

Sheet: /Sound mixer/
File: sound_mixer.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 14/17

Horizontal signals

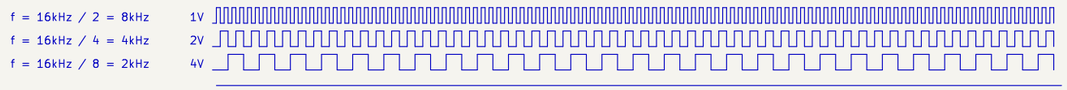


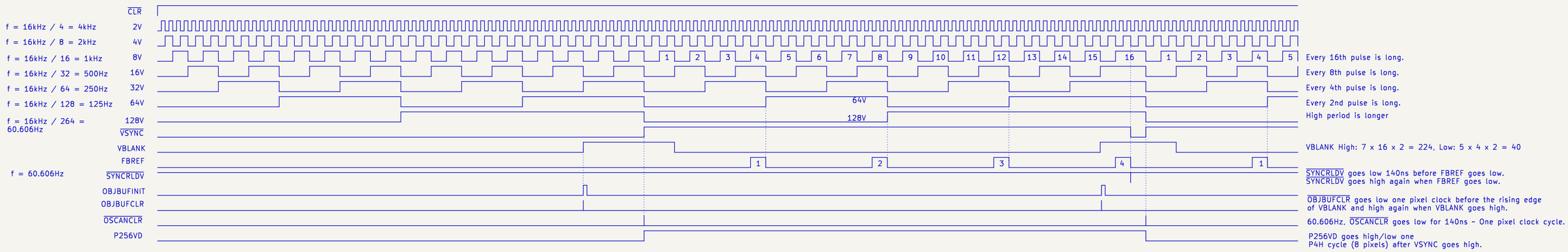
– The first VCNTEN is skipped after reset.
It goes low 140ns before HSYNC goes low,
and high again when HSYNC goes low.
VCNTEN is active right before every second falling edge of HSYNC.

– CPURES goes high, and stays high, on the seventh falling edge of HSYNC.

The first HSYNC is at HCOUNT 175 after reset.

Vertical signals





FBREF is measured at pin 22. At U186:3 FBREF is delayed by 22ns going high and 12ns going low.

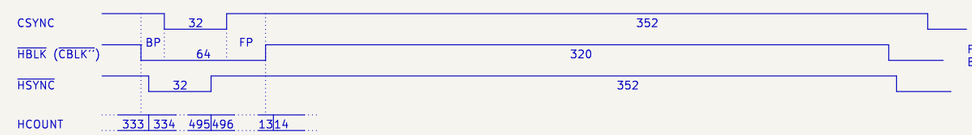
Horizontal and vertical synch timing diagrams

Pixel clock CLK2 = 18.432MHz / 3 = 6.144MHz

Horizontal lines
HSYNC freq = 6.144MHz / 384 = 16kHz
HSYNC period = 1 / 16kHz = 62.5us
Pixels / line: 384
Active pixels: 320
Blank pixels: 64

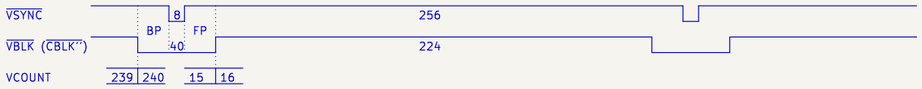
Vertical lines
VSYNC freq = 16kHz / 264 = 60.606060Hz
VSYNC period = 1 / f = 264 / 16kHz = 16.5ms
scanlines / frame: 264
Active lines: 224
Blank lines: 40

The numbers in the HSYNC and HBLK diagrams are HSYNC cycles.
All edges are synchronised to the rising edge of CLK2.



Front Porch (FP) 24 px,
Back Porch (BP) 8 px

The numbers in the VSYNC and VBLK diagrams are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.



Front Porch (FP) and
Back Porch (BP) 16 HSYNC cycles.

HCOUNT is bits [256H, 128H, 64H, 32H, 16H, 8H, P4H, P2H, P1H]
VCOUNT is bits [128V, 64V, 32V, 16V, 8V, 4V, 2V, 1V]

CBLK'' is at the output of color mixer 007327.
CBLK' is shifted inside the 007327 one pixel clock.

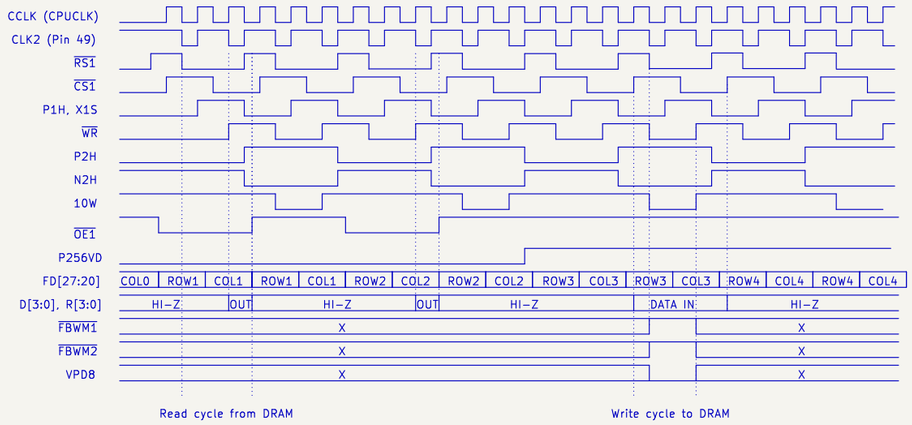
Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /Timing diagrams/
File: timing_diagrams.kicad_sch

Title: Twin 16

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The same row and column addresses are repeated twice.
One for read cycle and the other for write cycles.

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Konami GX687 / GX785

Sheet: /Sprite timing diagrams/
File: sprite_timing_diagrams.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
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D_F1A 74LS04
D_G1B 74LS32
D_F1B 74LS04
D_G1A 74LS32
D_E1 74LS174
D_G1C 74LS32
D_F1C 74LS04
D_F2D 74LS32
74LS32 D_E2A
D_G2 6264SL
D_I2 6264SL
D_F2C 74LS32
D_F2A 74LS32
D_F2B 74LS32
D_H2 6264SL
D_J2 6264SL
D_E2D 74LS32
D_E2B 74LS32
D_E2C 74LS32
74LS245 D_C1
74LS245 D_D1
74LS245 D_C2
74LS245 D_D2
CN1 Conn_02x20_Odd_Even
D_F1D 74LS04
D_F1E 74LS04
D_F1F 74LS04
D_G1D 74LS32
D_F1G 74LS04
D_U1 M51944BSL
D_R1 10k
D_D11 1S1588
D_D12 1S1588
D_BT1 3V
PWR_FLAG

DB''[12..8]
DB'[15..0]
AB'[10..1]
AB[10..1]
DB[15..0]

CN5 is on main board.
CN1 is on sub board PWB451803 and mates to CN5.

If VCC > 4.25V OUT is activated

VMEM is connected to VCC on the RAM IC's.

GC903
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Sheet: /Battery Powered RAM/
File: bpram.kicad_sch

Title: Twin 16

Size: A3 | Date: 2024-09-08 | **Rev:**
KiCad E.D.A. 8.0.5 | Id: 17/17